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Effects of plyometric training on match running performance in elite U20 male soccer players at moderate altitude

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Abstract

Background The effectiveness of plyometric training for improving isolated physical performance in soccer players is well established; however, its transfer to match running performance, particularly under chronic moderate-altitude conditions (~ 2000 m), remains unclear. This study examined whether supplementing regular soccer training with a structured plyometric training programme would enhance friendly match running performance in elite U20 male soccer players training at moderate altitude.

Methods Eighteen elite academy players completed a 16-week quasi-experimental, non-randomized, sequential within-subject intervention consisting of two consecutive 8-week training phases: regular soccer training only (control) followed by regular training supplemented with plyometric training (experimental). Match running performance was assessed during friendly matches using 10-Hz global positioning system technology (Catapult Sports, Australia) before the intervention, after the control phase, and after the plyometric training phase. Distance per minute (DPM), top speed (TS), high-speed running distance (HSR), sprint distance (SD), and explosive accelerations (EA) and decelerations (ED) were analyzed using generalized linear mixed models to account for repeated match observations within players. The fixed sequential design should be considered when interpreting the findings, as potential time-related and seasonal influences cannot be fully excluded.

Results Following the regular soccer training phase, trivial or unclear changes were observed in DPM (ES; ±95%CL: 0.02 ± 0.19), HSR (0.13 ± 0.43), SD (0.01 ± 0.55), EA (0.03 ± 0.34), and ED (0.13 ± 0.37), whereas TS showed a substantial increase (0.63 ± 0.12). Following the phase that included additional plyometric training, substantial increases were observed in DPM (1.66 ± 0.19), TS (1.66 ± 0.12), HSR (1.18 ± 0.40), SD (1.39 ± 0.51), EA (1.03 ± 0.33), and ED (1.15 ± 0.38).

Conclusion The findings indicate that adding plyometric training to regular soccer training may provide an additional stimulus associated with enhanced match running performance in elite youth soccer players, even under long-term moderate-altitude conditions. However, the results should be interpreted cautiously given the non-randomized sequential design.

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Keywords Youth athletes, Neuromuscular training, High-intensity activity, GPS monitoring, Adolescent, Soccer, Plyometric Exercise, Athletic Performance, Global Positioning System

Introduction

Soccer is considered a high-intensity intermittent team sport. Players are required to repeatedly perform walking, jogging, running, high-speed running, sprinting, accelerations, decelerations, and rapid changes of direction throughout match play [1, 2]. Early match-analysis research has demonstrated that these intermittent movement demands impose substantial physiological and neuromuscular stress on players and are closely associated with decisive match actions such as goal scoring, defensive interventions, and transitional play [3]. Consequently, match running performance has become a central construct in soccer performance analysis. Greater emphasis is now placed on high-intensity and explosive actions rather than total running distance alone [4, 5].

As performance analysis techniques have evolved, high-speed running, sprinting, and rapid changes in velocity have emerged as key differentiators between competitive standards and playing levels [6], despite soccer players only sprinting for 1–11% of total game time [2]. In elite youth soccer, match running performance has gained additional relevance, as physical capacities contribute not only to immediate match performance but also to long-term athlete development and progression toward senior professional levels [7]. Longitudinal and cross-sectional studies have shown that youth players who exhibit superior high-intensity running and sprint-related match outputs are more likely to transition successfully to higher competitive levels [8, 9]. Therefore, identifying training strategies capable of enhancing match running performance in elite youth populations is of considerable practical importance.

To enhance these high-intensity and explosive capacities, Plyometric training (PT) has been widely adopted in soccer to improve explosive performance [10, 11]. PT is considered highly relevant to soccer-specific actions, given the reliance on rapid force production in jumping, sprinting, and changes of direction [12–14]. PT is therefore understood to enhance neuromuscular performance through mechanisms such as elastic energy utilisation and stretch reflex activation [15]. Early experimental studies demonstrated that PT can improve sprint and jump performance in soccer players, and subsequent systematic reviews and meta-analyses confirmed these benefits across youth and adult populations [16, 17]. More recent meta-analytical evidence has refined this understanding, indicating that programme duration, weekly frequency, and cumulative training exposure are important moderators of sprint-related adaptations [18, 19]. In adolescent soccer players specifically, a 2025 systematic review with meta-analysis further confirmed that supplementing regular soccer training with PT leads to meaningful improvements in jumping, sprinting, and change-of-direction performance [20], outcomes typically assessed via controlled field- or laboratory-based tests.

Despite robust evidence for PT-induced improvements in isolated physical capacities, the extent to which these neuromuscular adaptations translate into match running performance remains less clear, because match running performance represents context-dependent locomotor output influenced by tactical and environmental constraints, whereas test metrics reflect isolated physical capacity in standardised conditions [21, 22]. The widespread adoption of global positioning system (GPS) technology has enabled objective quantification of match running performance, including speed-derived metrics and high-intensity acceleration and deceleration actions [23, 24].

In addition to tactical and contextual influences, environmental conditions such as altitude exposure may also affect the expression of high-intensity running performance during match play. Athletes who chronically reside and train at moderate altitude (1500–2500 m) demonstrate physiological and neuromuscular adaptations that differ from those observed at sea level, including altered oxygen availability, metabolic stress, and fatigue responses during repeated high-intensity efforts [25]. While altitude and hypoxic training strategies have been extensively examined in endurance sports, less attention has been given to how chronic moderate-altitude exposure may interact with neuromuscular training interventions such as PT in team-sport athletes [26]. Because high-intensity match actions rely heavily on repeated explosive efforts and recovery capacity, altitude-related physiological stress may theoretically influence the transfer of PT-induced neuromuscular adaptations to match running performance. However, most PT studies examining physical or match-related outcomes have been conducted at or near sea level. Consequently, it remains unclear whether PT-related improvements in match running performance are preserved, attenuated, or enhanced under chronic moderate-altitude conditions.

Accordingly, the purpose of this study was to examine whether the addition of a structured plyometric training programme to regular soccer training was associated with changes in friendly match running performance in elite U20 male soccer players who chronically reside and train at moderate altitude. This study extends previous PT research by combining GPS-derived match analysis

with a longitudinal training intervention conducted under chronic moderate-altitude conditions. Match running performance was assessed using GPS-derived metrics reflecting overall work rate as well as high-intensity and explosive actions. It was hypothesised that the inclusion of plyometric training would result in greater improvements in high-intensity running, sprint-related match outputs, and acceleration and deceleration actions compared with regular soccer training alone.

Method

Study design and participants

This study employed a 16-week quasi-experimental, non-randomized, sequential within-subject design with a fixed sequential intervention structure, as illustrated in Fig. 1, to determine whether the addition of plyometric training (PT) to regular soccer training would improve friendly match running performance in elite U20 male soccer players training at moderate altitude (~2000 m). The study consisted of three assessment time points: baseline (Pre), after the first 8-week phase (Post 1), and after the second 8-week phase (Post 2). As shown in Fig. 1, the first phase consisted of regular soccer training only (control phase), and the second phase consisted of regular soccer training plus additional plyometric training (PT phase). A 7-day recovery period separated the two phases. Given the fixed sequential design, potential influences of seasonal progression, accumulated training exposure, and maturation effects cannot be completely excluded. However, the inclusion of an initial control phase and standardized training and match conditions were intended to minimize these influences. The independent variable was training phase (Pre, Post 1, Post 2). The dependent variables were GPS-derived match running performance metrics: distance per minute (DPM), top speed (TS), high-speed running distance (HSR), sprint distance (SD), explosive accelerations (EA), and explosive decelerations (ED).

Eighteen elite U20 male outfield players participated (age: 18.4 ± 0.7 years; height: 176.4 ± 5.3 cm; body mass: 71.6 ± 6.2 kg; training experience: 9.1 ± 1.6 years). These players were full-time academy athletes competing within a professional club system and trained in a high-performance environment. The squad had resided and trained continuously at a moderate altitude (~ 2000 m) for an extended period prior to the commencement of the study and throughout the entire experimental duration, thereby minimizing the potential influence of acute altitude exposure on performance outcomes. All matches included in the analysis were friendly matches of comparable structure and importance, and were played against the same opponents at each time point. Player positions remained consistent across matches throughout the study. Players were included if they: [1] were registered members of the U20 squad throughout the observation period; [2] were free from musculoskeletal injuries that restricted full participation in training and match play; and [3] completed at least two full friendly matches (≥ 90 min of match exposure) at each testing time point. Players who sustained injuries requiring withdrawal from match participation or who failed to meet the minimum match exposure criteria were excluded from the final analysis. Following the application of the exclusion criteria, the final sample comprised n = 18. The positional distribution of the final sample was as follows: defenders (n = 4), midfielders (n = 7), and forwards (n = 7). To minimize measurement variability, all matches were conducted on the same natural grass pitch using a standardized match format consisting of two 45-minute halves. Matches at each testing time point were played against the same opponents, while player positional roles remained consistent throughout the study period. Environmental conditions were also standardized across matches, with ambient temperature maintained between 18 and 22 °C and relative humidity between 40 and 55%. In addition, to reduce inter-unit variability, each player wore the same 10-Hz GPS device throughout the study, and all devices were activated at least 15 min prior to kickoff to ensure stable satellite signal acquisition. The study was conducted in accordance with the Declaration

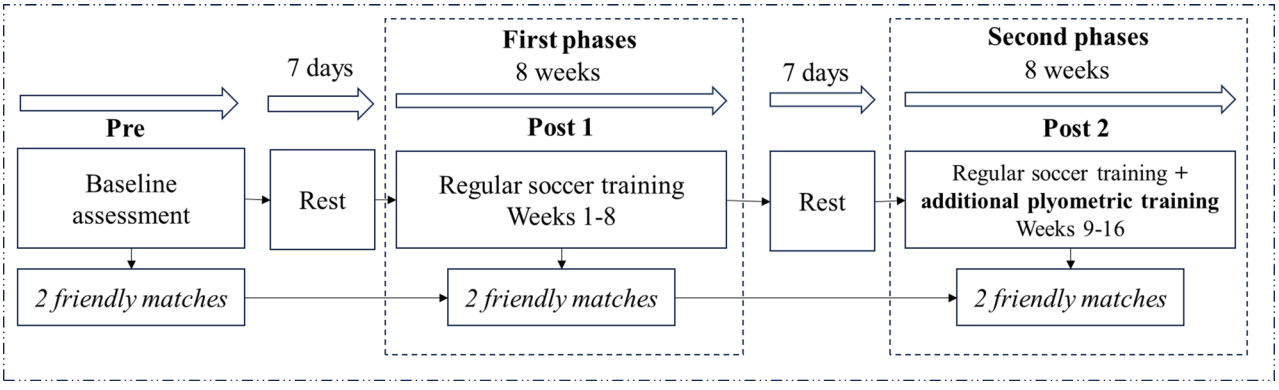


Fig. 1 Summary of study design and experimental procedures

of Helsinki and was approved by the institutional ethics committee. Players and their legal guardians were fully informed of the study procedures, potential risks, and benefits, and written informed consent was obtained prior to participation.

Intervention protocol

The experimental timeline is summarized in Fig. 1. The intervention consisted of two consecutive 8-week phases. During Phase 1 (Weeks 1–8), players completed regular soccer training only, which served as the control phase. During Phase 2 (Weeks 9–16), players continued the same regular soccer training programme with the addition of structured plyometric training (PT phase). Regular soccer training during both phases consisted of technical–tactical drills, small-sided games, positional play exercises, aerobic and anaerobic conditioning, and friendly match preparation. Weekly training frequency was maintained at five to six field-based sessions and one friendly match per week throughout the study period. Training content, coaching staff, and overall team schedule remained consistent across both phases, with the addition of plyometric training representing the primary difference during Phase 2. The plyometric training programme [27], detailed in Table 1, was designed according to the FITT principle. Training was performed twice per week (frequency) and consisted of progressive moderate-to-high intensity explosive exercises (intensity), lasting approximately 20–25 min per session (time). The exercises included bilateral and unilateral hopping, countermovement jumps, horizontal hops, bounding, and sprint drills (type). Training volume progressed systematically across the 8-week PT phase. Total ground contacts per session (excluding sprinting) increased from 80 contacts in Week 1 to a peak of 190 contacts in Week 7. Week 8 was implemented as a tapering week (60 contacts) to reduce neuromuscular fatigue prior to post-intervention testing. Sprint volume progressed from 2 × 20 m in Week 1 to 3 × 50 m in Week 7, before being reduced in Week 8 as part of the taper.

Measurement protocols

Match running performance was assessed during friendly matches, as illustrated in Fig. 1. Two friendly matches were played at each testing time point (Pre, Post 1, and Post 2) against the same opponents to enhance consistency and reduce contextual variability. Matches were conducted on regulation-size natural grass pitches under standardized environmental conditions (ambient temperature 18–22 °C; relative humidity 40–55%). Only official match periods (two 45-minute halves) were included in the analysis. All matches were monitored using 10-Hz global positioning system (GPS) units (Catapult Sports, Australia). The 10-Hz GPS system has demonstrated acceptable validity and reliability for measuring total distance, high-speed running, and acceleration-related metrics in team-sport athletes (intraclass correlation coefficients > 0.85; coefficient of variation < 5%) [28, 29]. To minimize inter-unit variability, each player wore the same GPS device throughout the study. Devices were activated at least 15 min prior to kickoff to ensure optimal satellite signal acquisition. GPS data were downloaded and processed using the manufacturer's proprietary software. Raw data were visually inspected for signal dropouts and obvious tracking errors prior to analysis. The following variables were extracted to quantify match running performance: distance per minute (DPM), top speed (TS), high-speed running distance (HSR), sprint distance (SD), explosive accelerations (EA), and explosive decelerations (ED). Only full-match observations (≥ 90 min) with valid GPS recordings were included in the final analysis.

Statistical analysis

Generalized linear mixed models were realised using PROC GLIMMIX in SAS® OnDemand for Academics to examine the effects of training phase on match running performance in elite U20 male soccer players chronically trained at moderate altitude. Separate models were constructed for each match running performance variable, which served as the dependent variable in the analyses. In

Table 1 Plyometric training programme (sets and repetitions) performed over the 8-week intervention period [27]

Exercise	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Submaximal bilateral hopping	2 × 20	2 × 10	2 × 10	3 × 10	2 × 20	3 × 20	3 × 20	2 × 20
Maximal bilateral hopping	5 × 6	2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	3 × 10	2 × 5
Countermovement jumps	1 × 10	2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	1 × 10
Unilateral vertical hopping		2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	
Bilateral horizontal hops		2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	
Unilateral horizontal hops			2 × 10	2 × 10	2 × 10	2 × 10	2 × 10	
Bounding					2 × 10	2 × 10	2 × 10	
Sprinting	2 × 20 m	3 × 10 m	2 × 15 m	2 × 20 m	2 × 30 m	2 × 50 m	3 × 50 m	2 × 20 m
Total foot contacts per leg*	80	100	120	130	160	180	190	60

* Total number of foot contacts excluding sprinting

The plyometric training programme [27]

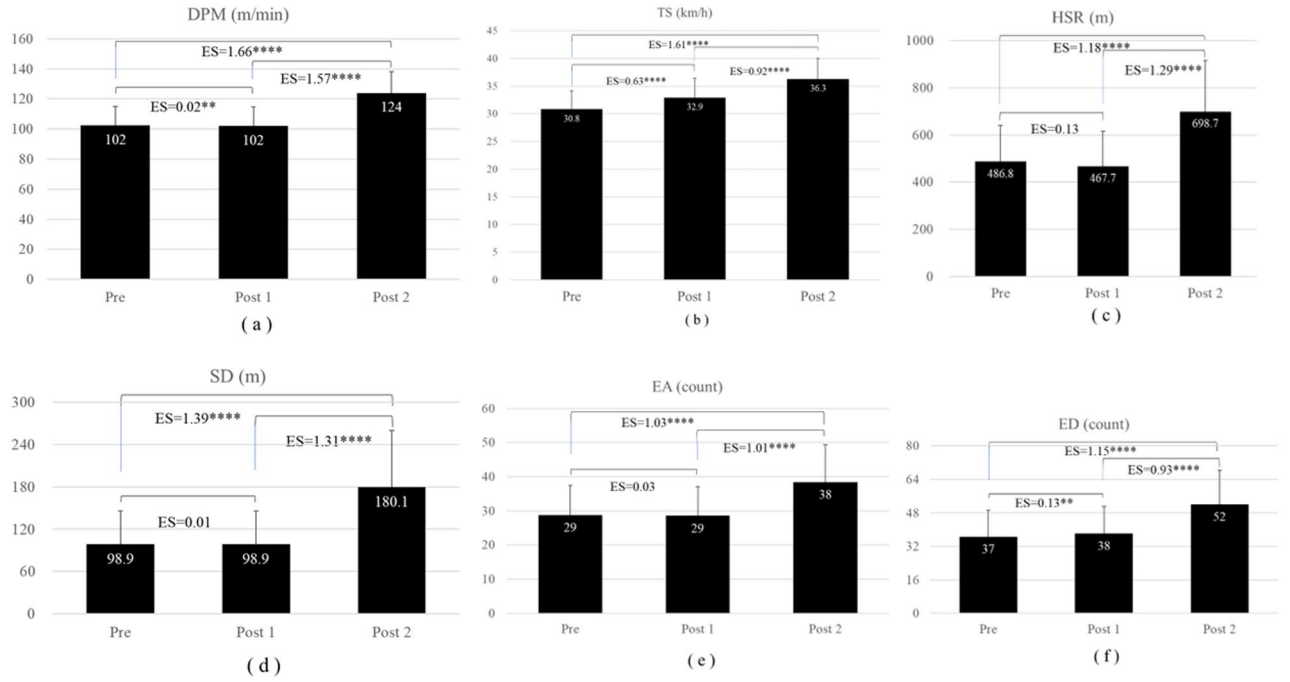


Fig. 2 Changes in running performance variables across different training phases during the preparatory period. **a** DPM (m/min); **b** TS (km/h); **c** HSR (m); **d** SD (m); **e** EA (count); **f** ED (count). Number of asterisks indicate the likelihood for the magnitude of the true change as follows: *possible; **likely; ***very likely; ****most likely

each model, training phase (Pre, Post 1, and Post 2) was included as a fixed effect. A random intercept for player was specified to account for repeated match observations within individuals across multiple friendly matches.

The models estimated the mean value of each dependent variable at each training phase. Effects of the different training phases on match running performance were quantified and expressed as standardized effect sizes (ES), calculated from the model-based mean changes between training phases. Effect sizes were interpreted using the following qualitative thresholds: trivial (< 0.2), small (0.2–0.6), moderate (0.6–1.2), large (1.2–2.0), and very large (> 2.0). Uncertainty in the estimated effects was assessed using 95% confidence intervals (CI) [30]. Clear effects were further described using a magnitude-based qualitative likelihood that the true effect was substantial or trivial (whichever probability was greater), according to the following scale: <0.5% most unlikely, 0.5–5% very unlikely, 5–25% unlikely, 25–75% possibly, 75–95% likely, 95–99.5% very likely, and > 99.5% most likely. In the present study, only effects with a likelihood of ≥ 95% were included in the interpretation and discussion.

Results

Values of DPM, TS, HSR, SD, EA, and ED during friendly matches at Pre, Post 1, and Post 2 are presented in Fig. 2.

Effects of the regular soccer training phase (pre to post 1)

Following the regular soccer training phase, trivial or unclear changes were observed in DPM (ES; ±95%CL, 0.02; ±0.19), HSR (0.13; ±0.43), SD (0.01; ±0.55), EA (0.03; ±0.34), and ED (0.13; ±0.37), whereas TS showed a substantial increase (0.63; ±0.12).

Effects of the plyometric training phase (post 1 to post 2)

Following the phase that included additional plyometric training, substantial increases were observed in DPM (1.66; ±0.19), TS (1.66; ±0.12), HSR (1.18; ±0.40), SD (1.39; ±0.51), EA (1.03; ±0.33), and ED (1.15; ±0.38).

Comparison between training phases

Compared with the regular soccer training phase, DPM (1.57; ±0.18), TS (0.92; ±0.12), HSR (1.29; ±0.38), SD (1.31; ±0.46), EA (1.01; ±0.31), and ED (0.93; ±0.33) were substantially greater following the plyometric training phase.

Discussion

The present study aimed to determine whether supplementing regular soccer training with a structured plyometric training (PT) programme would enhance friendly match running performance in elite U20 players chronically training at moderate altitude. The main findings were that eight weeks of regular soccer training alone produced largely trivial or unclear changes in GPS-derived match running variables, whereas the

subsequent addition of PT was associated with improvements in overall work rate, top speed, high-speed running distance, sprint distance, and explosive acceleration and deceleration actions. Importantly, these improvements were expressed within an ecologically valid match context. Given that gains observed in isolated sprint or jump tests do not necessarily translate to competitive performance, the present findings provide novel evidence that PT-induced adaptations may meaningfully transfer to match play in elite youth soccer players training at altitude.

The present results are broadly consistent with the extensive literature demonstrating that PT improves sprinting, jumping, and change-of-direction performance in soccer players across age groups and competitive standards [16, 17, 31–33]. However, rather than reiterating the well-established effectiveness of PT in isolated physical tests, the key contribution of the present study lies in demonstrating that these adaptations can be expressed in match-derived running performance. More recent meta-analyses have further refined this evidence base, highlighting that PT-induced adaptations are influenced by programme duration, weekly frequency, and cumulative training exposure, with youth and young adult players often exhibiting pronounced responsiveness to relatively low-volume interventions [18, 34]. Most previous studies have relied on isolated physical performance tests, and relatively few have examined whether PT-related neuromuscular adaptations translate into meaningful changes during match play. Given that match running performance is constrained by tactical roles, opposition behaviour, and contextual factors, improvements in physical capacity do not necessarily result in increased match outputs [21, 22]. By directly assessing match running performance using GPS during standardised friendly matches, the present study extends the literature by confirming that PT adaptations can be observed under match-specific constraints.

From a physiological and mechanical perspective, the most pronounced improvements following the PT phase were observed in high-speed running, sprint distance, and explosive acceleration and deceleration actions, whereas changes during the control phase were minimal. This pattern aligns with contemporary match analysis research indicating that high-intensity and explosive actions, rather than total running distance alone, are key determinants of competitive level and match impact in modern soccer [5, 6]. PT is known to enhance rate of force development, neuromuscular coordination, and musculotendinous stiffness, which underpin rapid force production during short ground contact times [35, 36]. Sprint performance is influenced by both slow (acceleration phase) and fast (brief ground contact times) components of the stretch-shortening cycle (SSC), alongside biomechanical factors such as stride length and frequency, horizontal force production, and lower-body strength and power [37–39]. Given that sprinting may represent a highly specific manifestation of plyometric function [40], the inclusion of sprint drills within the PT programme likely contributed to improvements in horizontal force application and acceleration capacity. Accelerations and decelerations impose substantial eccentric and concentric neuromuscular demands. Improvements in these qualities may enable players to achieve higher velocities more frequently and tolerate repeated mechanical loading during match play, thereby increasing the expression of high-intensity actions. Additionally, the increase in overall work rate following the PT phase may reflect a reduced relative neuromuscular cost of high-intensity actions, enabling players to sustain greater activity levels across the duration of a match.

In contrast, the trivial changes observed during the control phase alone align with longitudinal observations in elite youth and senior players, where habitual exposure to technical–tactical training may be insufficient to further stimulate neuromuscular adaptations without targeted strength and power-oriented stimuli [22, 41]. A novel contextual aspect of the present study is the chronic moderate-altitude training background of the participants. Acute hypoxic exposure has been shown to impair repeated sprint ability and high-intensity running performance, whereas long-term residence at moderate altitude results in more stable adaptations related to oxygen transport and fatigue tolerance [25]. Nevertheless, little is known regarding how chronic altitude exposure interacts with neuromuscular training modalities in team-sport athletes.

The present findings suggest that PT may remain an effective stimulus for enhancing match running performance under chronic moderate-altitude conditions. However, these interpretations should be considered cautiously, as the present design does not allow definitive conclusions regarding the mechanistic interaction between altitude exposure and PT-induced adaptations. PT-induced adaptations are predominantly neural and mechanical in nature and may therefore be less constrained by oxygen availability than aerobic or metabolic adaptations [16, 31]. Accordingly, the present findings should be interpreted as preliminary evidence supporting the potential compatibility of PT with chronic moderate-altitude training environments, rather than confirmation of a specific altitude-related mechanism. Thus, in altitude-based training environments, PT may serve as a complementary strategy targeting performance determinants not fully addressed by soccer-specific or altitude-related physiological adaptations alone.

From an applied perspective, the present findings have important implications for elite youth soccer

environments, particularly those based at altitude or in hypoxic regions. The limited changes observed during the control phase suggest that technical–tactical training may not provide a sufficient stimulus to further enhance high-intensity match running performance in well-trained youth players. In contrast, the addition of structured PT appears to be a time-efficient strategy to augment match-related physical outputs, particularly those associated with decisive actions such as sprinting, pressing, and rapid transitions. However, increases in explosive accelerations and decelerations also imply greater mechanical loading, which has been linked to muscle injury risk when not appropriately managed. Accordingly, the integration of PT within elite youth programmes should be accompanied by careful load monitoring and recovery strategies to maximise performance benefits while minimising injury risk [24]. In addition, practitioners should consider the potential monitoring burden associated with integrating additional PT sessions alongside regular soccer training, particularly during periods of congested schedules or elevated cumulative training loads. Careful management of neuromuscular fatigue and recovery may therefore be necessary to minimise excessive mechanical stress and optimise training adaptation. In support of the broader sport-specific relevance of plyometric training, a recent pilot study in youth goalkeepers reported improvements in penalty-related ball-blocking performance following a structured plyometric programme [42]. Although the performance outcomes and population differ from the present GPS-derived running metrics in outfield players, these findings collectively suggest that plyometric training may enhance multiple match-relevant capacities beyond isolated field tests.

Several methodological limitations should be considered. First, the study employed a non-randomized, sequential design with a fixed intervention order, which does not completely exclude potential time or seasonal effects, although the trivial changes observed during the control phase provide some internal support for the interpretation that the subsequent improvements may have been associated, at least in part, with the PT intervention. However, alternative explanations such as increased familiarity with match demands and testing procedures, natural seasonal progression, and contextual match factors cannot be excluded. In addition, the absence of a parallel randomized control group limits the ability to establish causal inference regarding the isolated effects of the plyometric intervention. Second, match performance was assessed during friendly matches rather than official competition, and although this allowed standardisation, psychological and tactical demands may differ from competitive fixtures. In addition, matches were played repeatedly against the same opponents under similar conditions, which may have influenced the observed performance responses. Furthermore, while 10-Hz GPS devices demonstrate acceptable validity, measurement error increases for high-acceleration and deceleration actions. Finally, the relatively small sample size ($n = 18$), combined with the absence of an a priori sample size calculation, may limit statistical power and the generalisability of the findings, although the sample size is consistent with similar longitudinal intervention studies in elite youth soccer.

Future investigations should aim to integrate internal load measures, neuromuscular assessments, and biomechanical profiling to further elucidate the mechanisms underpinning PT-induced changes in match performance. In particular, comprehensive assessment of stretch–shortening cycle (SSC) function may provide deeper insight into the neuromechanical adaptations associated with improved sprint and acceleration performance. While sprint performance reflects aspects of fast SSC function, additional testing—such as countermovement jump (CMJ), reactive strength index (RSI), and leg stiffness assessments—would enable differentiation between slow and fast SSC components and provide a more complete neuromuscular profile of soccer players. Such measures may also inform monitoring of fatigue, training readiness, and injury risk, and help clarify the relationship between SSC characteristics and sprint performance in match contexts. Accordingly, future studies should explore the development of multiple SSC components in male soccer players and examine their direct association with match-derived running metrics.

Conclusion

This study suggests that adding a structured plyometric training programme to regular soccer training may enhance friendly match running performance in elite U20 male soccer players chronically training at moderate altitude. The inclusion of plyometric training was associated with greater improvements in overall match activity and high-intensity running actions compared with regular soccer training alone, which may reflect the transfer of neuromuscular adaptations to match play. These findings provide ecologically valid evidence that plyometric training may represent a potentially useful complementary training strategy for elite youth players under long-term moderate-altitude conditions. Practitioners may therefore consider integrating plyometric training within youth soccer programmes to support high-intensity match-related actions, while ensuring appropriate load management.

Abbreviations

PT	Plyometric training
DPM	distance per minute
TS	Top speed

HSR High-speed running distance
SD Sprint distance
EA Explosive accelerations
ED Explosive decelerations

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Authors' contributions
LF was involved in data collection, analysis and interpretation, drafted and revised the manuscript. LM, XB, XZ, YG, ZF, YJ, HW, HL drafted and revised the manuscript. All the authors approved the final version of the manuscript.

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Data availability
The datasets generated and analysed during the current study are provided in the supplementary files accompanying this article.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with the Declaration of Helsinki. The study protocol was reviewed and approved by the Ethics Committee of the School of Physical Education & Sports Science, South China Normal University. Written informed consent was obtained from all participants prior to their inclusion in the study.

Consent for publication
Not applicable. This manuscript does not contain any identifiable individual data, images, or personal information that would require consent for publication.

Competing interests
The authors declare no competing interests.

Clinical trial number
Not applicable.

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